47202 - Introduction to Future Energy - 2026

Version. 25 August 2026

Tuesdays 13-17
Lectures 13-15: building 306, Auditorium 33
Activities 15-17: building 306, rooms 108A, 108B, 196, 197, 198, 199
Texts for preparation on DTU Learn

#	Title	Content	Teacher
Introduction			
1 (1/9)	Introduction	Welcome and outline of the course. Basic principles for energy.	Oluf
2 (8/9)	Thermodynamics and thermal energy	The need to know thermodynamics for energy technology. Thermal engines, heat pumps and geothermal energy.	Oluf
Energy sources			
3 (15/9)	Solar	Solar resource, solar thermal energy, and photovoltaics.	Rafael
4 (22/9)	Wind	Wind resource and wind turbines (other mechanical systems in brief)	Rafael
5 (29/9)	Biomass	Biomass resource, thermochemical conversion, and biofuels. (other bio-e-fuel concepts in brief)	Rafael
6 (6/10)	Nuclear	Fission and fusion.	Oluf
Autumn holiday			
Energy conversion, transmission and storage			
7 (20/10)	Electrochemical cells	Basics of electrochemisty and electrochemical cells.	Oluf
8 (27/10)	Hydrogen and Power-to-X	Fuel cells, electrolyzers and hydrogen as an energy carrier. Electrofuels.	Oluf
9 (3/11)	Infrastructure	Electricity transportation principles and the energy grid (other energy integration infrastructure in brief)	Rafael
10 (10/11)	Energy storage	The different technologies. Choice and scale.	Oluf
11 (17/11)	Batteries	Batteries and flow batteries.	Oluf
Wrapping up			
12 (24/11)	Energy economics	Basic parameters and trends in energy economics	Rafael
13 (1/12)	Wrap-up	Missing bits and pieces. Wrap-up.	Oluf
15/12	Exam (E4A)	Multiple choice 2 hours (*time of the day pending*)	

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